A Linear Algebra Guide to the Strange Life of High-Dimensional Data

Preliminary Working Report — Questions 1 & 2

K. Vishnu Kumar Adithya Kesan Jayakanth Adithya Anand

26 May 2026

Internal draft: full proofs and all details. Questions 1 and 2 only; to be consolidated with Question 3 into the 10-page submission later.

Scope and status of this draft

This preliminary report contains the *complete* worked solutions to **Question 1** (Linear Algebra of Markov Matrices) and **Question 2** (The Power Method and Singular Value Decomposition), with every proof written out in full. It is a working document, not the final submission: the generic report scaffolding (literature, data provenance, sensitivity studies) is omitted because Questions 1 and 2 are purely analytical, and Question 3 (the data analysis on `mystery_matrix1.npy`) will be merged in later before the work is compressed to the 10-page cap.

Coloured tags mark every place this document departs from the team's handwritten notes. Each tag has a matching entry in `extras.md`:

- **[MISS: ...]** — a part of the problem not yet attempted in the notes.
- **[CORRECTED: ...]** — a mathematical error in the notes, corrected here.
- **[EXTRA: ...]** — content added beyond the notes (an elaboration or a missing explanation).

Notation and conventions

$\mathbf{M}$, $\mathbf{X}$	matrices (bold uppercase).
$\mathbf{v}, \mathbf{u}, \mathbf{x}, \mathbf{p}$	column vectors (bold lowercase).
$\mathbf{X}^\top$	transpose of $\mathbf{X}$, $(\mathbf{X}^\top)_{ij} = \mathbf{X}_{ji}$.
λ_i, $\mathbf{v}_i$	eigenvalue / eigenvector pair, ordered $\lvert\lambda_1\rvert \geq \lvert\lambda_2\rvert \geq \cdots$.
σ_i	singular value, $\sigma_i = \sqrt{\lambda_i}$.
$\lVert\mathbf{x}\rVert$	Euclidean norm, $\lVert\mathbf{x}\rVert = \sqrt{\mathbf{x}^\top\mathbf{x}}$.

A set of vectors is *orthonormal* if $\mathbf{v}_i^\top \mathbf{v}_j = 0$ for $i \neq j$ and $\mathbf{v}_i^\top \mathbf{v}_i = 1$. We use the **spectral theorem** as a given: every real symmetric matrix has real eigenvalues and an orthonormal basis of real eigenvectors.

1 Linear Algebra of Markov Matrices

Setup. Two organism types have fractional populations $p_1(t), p_2(t)$, evolving by $\mathbf{p}(t+1) = \mathbf{M}\,\mathbf{p}(t)$ with

$$\mathbf{p}(t) = \begin{pmatrix} p_1(t) \\ p_2(t) \end{pmatrix}, \qquad \mathbf{M} = \begin{pmatrix} 1-\varepsilon & 0 \\ \varepsilon & 1 \end{pmatrix}, \qquad \varepsilon \in (0,1) \text{ fixed.}$$

Componentwise this is

$$p_1(t+1) = (1-\varepsilon)\,p_1(t), \qquad p_2(t+1) = \varepsilon\,p_1(t) + p_2(t). \tag{1}$$

Each column of $\mathbf{M}$ sums to 1 (it is *column-stochastic*); a fraction ε of type-1 converts to type-2 each generation.

1.1 (a) Conservation

Claim. $p_1(t) + p_2(t)$ is constant in t.

Proof. Let $S(t) = p_1(t) + p_2(t)$. Adding the two equations in (**??**),

$$S(t+1) = p_1(t+1) + p_2(t+1) = (1-\varepsilon)p_1(t) + \big(\varepsilon\,p_1(t) + p_2(t)\big) = p_1(t) + p_2(t) = S(t).$$

So $S(t+1) = S(t)$ for every t. By induction $S(t) = S(0) = p_1(0) + p_2(0)$, a constant independent of t. □

Equivalent one-line argument. Because every column of $\mathbf{M}$ sums to 1, the row vector $\mathbf{1}^\top = (1,1)$ satisfies $\mathbf{1}^\top\mathbf{M} = \mathbf{1}^\top$, hence $\mathbf{1}^\top\mathbf{p}(t+1) = \mathbf{1}^\top\mathbf{M}\,\mathbf{p}(t) = \mathbf{1}^\top\mathbf{p}(t)$, i.e. $S(t+1) = S(t)$.

1.2 (b) Eigenspectrum

The eigenvalues solve $\det(\mathbf{M} - \lambda\mathbf{I}) = 0$:

$$\det\begin{pmatrix} 1-\varepsilon-\lambda & 0 \\ \varepsilon & 1-\lambda \end{pmatrix} = (1-\varepsilon-\lambda)(1-\lambda) = 0 \quad \Longrightarrow \quad \boxed{\lambda_1 = 1, \qquad \lambda_2 = 1-\varepsilon.}$$

(The matrix is lower-triangular, so the eigenvalues are simply its diagonal entries.) The eigenvectors come from $(\mathbf{M} - \lambda_i\mathbf{I})\mathbf{v}_i = \mathbf{0}$.

For $\lambda_1 = 1$: $\begin{pmatrix} -\varepsilon & 0 \\ \varepsilon & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0} \Rightarrow -\varepsilon x = 0 \Rightarrow x = 0$, y free. Take $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

For $\lambda_2 = 1-\varepsilon$: $\begin{pmatrix} 0 & 0 \\ \varepsilon & \varepsilon \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{0} \Rightarrow \varepsilon(x+y) = 0 \Rightarrow y = -x$. Take $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

1.3 (c) Linear combination

Assume $p_1(0) = \frac{1}{2}$. Write $\mathbf{p}(0) = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = c_1\begin{pmatrix} 0 \\ 1 \end{pmatrix} + c_2\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} c_2 \\ c_1 - c_2 \end{pmatrix}$. Matching components,

$$c_2 = p_1(0) = \tfrac{1}{2}, \qquad c_1 - c_2 = p_2(0) \;\Rightarrow\; c_1 = p_2(0) + \tfrac{1}{2}.$$

[CORRECTED: notes assumed $p_2(0) = \frac{1}{2}$ without stating it.] If, as is natural for fractional populations, the totals are normalised so that $p_1(0) + p_2(0) = 1$, then $p_2(0) = \frac{1}{2}$ and $c_1 = 1$, giving

$$\boxed{\mathbf{p}(0) = \mathbf{v}_1 + \tfrac{1}{2}\mathbf{v}_2.}$$

More generally $c_1 = p_2(0) + \frac{1}{2}$ and $c_2 = \frac{1}{2}$; note $c_1 = p_1(0) + p_2(0) = S(0)$, the conserved total from part (a).

1.4 (d) Eigenvectors make time evolution easy

With $\mathbf{p}(0) = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, apply $\mathbf{M}$ repeatedly. Since $\mathbf{M}\mathbf{v}_i = \lambda_i\mathbf{v}_i$, we have $\mathbf{M}^t\mathbf{v}_i = \lambda_i^t\mathbf{v}_i$, so

$$\mathbf{p}(t) = \mathbf{M}^t\mathbf{p}(0) = c_1\lambda_1^t\mathbf{v}_1 + c_2\lambda_2^t\mathbf{v}_2 \quad\Longrightarrow\quad \boxed{\mathbf{p}(t) = c_1\,\mathbf{v}_1 + c_2\,(1-\varepsilon)^t\,\mathbf{v}_2.}$$

The eigenbasis turns repeated matrix multiplication into scalar powers: each mode evolves independently by its own factor λ_i^t.

1.5 (e) Long-time behaviour

(i) Because $\varepsilon \in (0,1)$ we have $0 < 1-\varepsilon < 1$, so $(1-\varepsilon)^t \to 0$ as $t \to \infty$. Hence

$$\mathbf{p}(t) \xrightarrow{t\to\infty} c_1\mathbf{v}_1,$$

the eigenvector with the larger eigenvalue $\lambda_1 = 1 > \lambda_2$.

(ii) The structure of $\mathbf{M}$ explains this. The second column $(0,1)^\top$ means type-2 is *absorbing*: nothing leaves it. Type-1 leaks into type-2 at rate ε every step and never receives anything back. So all mass drains into type-2. Quantitatively the limit is

$$c_1\mathbf{v}_1 = \begin{pmatrix} 0 \\ c_1 \end{pmatrix} = \begin{pmatrix} 0 \\ p_1(0) + p_2(0) \end{pmatrix},$$

i.e. the entire conserved population $S(0)$ ends up as type-2. The eigenvalue $\lambda_1 = 1$ carries the conserved steady state, while $\lambda_2 = 1-\varepsilon < 1$ governs the decaying transient.

1.6 (f) Validation

[MISS: not attempted in the notes.] The closed form $\mathbf{p}(t) = c_1\mathbf{v}_1 + c_2(1-\varepsilon)^t\mathbf{v}_2$ should be checked against a direct numerical iteration of $\mathbf{p}(t+1) = \mathbf{M}\mathbf{p}(t)$ from $\mathbf{p}(0) = (\frac{1}{2}, \frac{1}{2})$, for a sample ε (e.g. 0.1), confirming agreement at every t and the limit $\mathbf{p}(\infty) = (0,1)$. *To be added in* `src/task1.py` *with a convergence plot; no numbers are invented here.*

1.7 (g) Time-varying conversion rate

Now $\varepsilon(t) = \varepsilon_0 e^{-t/t_0}$ with $\varepsilon_0, t_0 > 0$, so the update matrix changes each step: $\mathbf{M}(t) = \left(\begin{smallmatrix} 1-\varepsilon(t) & 0 \\ \varepsilon(t) & 1 \end{smallmatrix}\right)$.

[CORRECTED: the notes wrote $\mathbf{p}(t) = 1^t c_1 + \big(1-\varepsilon(t)\big)^t c_2$, which is not correct for a time-varying rate.] The key observation is that the eigenvectors $\mathbf{v}_1 = (0,1)^\top$ and $\mathbf{v}_2 = (1,-1)^\top$ *do not depend on* ε, so they are shared by *every* matrix $\mathbf{M}(t)$:

$$\mathbf{M}(t)\,\mathbf{v}_1 = 1\cdot\mathbf{v}_1, \qquad \mathbf{M}(t)\,\mathbf{v}_2 = \big(1-\varepsilon(t)\big)\mathbf{v}_2.$$

The state is the ordered product $\mathbf{p}(t) = \mathbf{M}(t-1)\mathbf{M}(t-2)\cdots\mathbf{M}(0)\,\mathbf{p}(0)$. Acting on $\mathbf{p}(0) = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$, the $\mathbf{v}_1$ mode is multiplied by 1 at every step, while the $\mathbf{v}_2$ mode picks up one factor $\big(1-\varepsilon(s)\big)$ per step. Therefore

$$\boxed{\mathbf{p}(t) = c_1\,\mathbf{v}_1 + c_2\Big[\textstyle\prod_{s=0}^{t-1}\big(1-\varepsilon_0 e^{-s/t_0}\big)\Big]\mathbf{v}_2.}$$

So the constant-rate factor $(1-\varepsilon)^t$ of part (d) is replaced by the **product** $\prod_{s=0}^{t-1}\big(1-\varepsilon(s)\big)$, not by $\big(1-\varepsilon(t)\big)^t$.

[EXTRA: qualitative consequence (beyond the notes).] Since $\sum_{s\ge 0}\varepsilon_0 e^{-s/t_0} < \infty$, the infinite product converges to a *strictly positive* limit $\Pi_\infty = \prod_{s=0}^{\infty}\big(1-\varepsilon_0 e^{-s/t_0}\big)$. Unlike the constant-$\varepsilon$ case, the $\mathbf{v}_2$ transient does *not* decay to zero: the conversion "switches off" as $\varepsilon(t)\to 0$, freezing a residual amount $c_2\Pi_\infty$ in the $\mathbf{v}_2$ direction. The system therefore settles at $\mathbf{p}(\infty) = c_1\mathbf{v}_1 + c_2\Pi_\infty\mathbf{v}_2$, i.e. type-1 is *not* fully depleted.

1.8 (h) Validate again

[MISS: not attempted in the notes.] The corrected product formula for part (g) should be validated the same way as (f): iterate $\mathbf{p}(t+1) = \mathbf{M}(t)\mathbf{p}(t)$ with $\varepsilon(t) = \varepsilon_0 e^{-t/t_0}$ and compare against $c_1\mathbf{v}_1 + c_2\prod_{s<t}(1-\varepsilon(s))\,\mathbf{v}_2$, checking the residual $c_2\Pi_\infty$ in the limit. *To be added in* `src/task1.py`.

2 The Power Method and Singular Value Decomposition

Setup. $\mathbf{X}$ is a symmetric $n\times n$ matrix with eigenvalues $|\lambda_1| > |\lambda_2| \ge \cdots \ge |\lambda_n|$ and an orthonormal eigenbasis $\mathbf{v}_1,\dots,\mathbf{v}_n$. The Power Method starts from a unit vector $\mathbf{x}_0$ and iterates

$$\mathbf{x}_{k+1} = \frac{\mathbf{X}\,\mathbf{x}_k}{\|\mathbf{X}\,\mathbf{x}_k\|}, \qquad k = 0,1,2,\dots \tag{2}$$

2.1 (a) The Power Method converges

Claim. If $\mathbf{x}_0$ has a nonzero component along $\mathbf{v}_1$, then $\mathbf{x}_k \to \pm\mathbf{v}_1$.

Proof. Normalisation in (**??**) only rescales, so $\mathbf{x}_k$ has the same *direction* as the unnormalised iterate $\mathbf{y}_k := \mathbf{X}^k\mathbf{x}_0$. Expand $\mathbf{x}_0$ in the orthonormal eigenbasis,

$$\mathbf{x}_0 = \sum_{i=1}^{n} a_i\mathbf{v}_i, \qquad a_i = \mathbf{x}_0^\top\mathbf{v}_i, \qquad a_1 \ne 0.$$

Since $\mathbf{X}^k\mathbf{v}_i = \lambda_i^k\mathbf{v}_i$,

$$\mathbf{y}_k = \mathbf{X}^k\mathbf{x}_0 = \sum_{i=1}^{n} a_i\lambda_i^k\mathbf{v}_i = \underbrace{a_1\lambda_1^k\mathbf{v}_1}_{\text{parallel to }\mathbf{v}_1} + \underbrace{\sum_{i=2}^{n} a_i\lambda_i^k\mathbf{v}_i}_{=\,\mathbf{r}_k\ \text{(residual)}}.$$

By orthonormality the residual length is $\|\mathbf{r}_k\|^2 = \sum_{i=2}^{n} a_i^2 \lambda_i^{2k}$. Measure the residual relative to the parallel part:

$$\rho_k := \frac{\|\mathbf{r}_k\|}{|a_1|\,|\lambda_1|^k} = \frac{1}{|a_1|}\sqrt{\sum_{i=2}^{n} a_i^2 \left(\frac{\lambda_i}{\lambda_1}\right)^{2k}}.$$

For every $i \geq 2$ we have $|\lambda_i/\lambda_1| < 1$, so each term $(\lambda_i/\lambda_1)^{2k} \to 0$ and hence $\rho_k \to 0$. Thus $\mathbf{y}_k$ aligns with $\mathbf{v}_1$, and after normalisation $\mathbf{x}_k \to \pm\mathbf{v}_1$ (the sign tracks $\operatorname{sign}(a_1\lambda_1^k)$). □

2.2 (b) Convergence rate and the spectral gap

Claim. The residual ratio shrinks each step by at least $|\lambda_2/\lambda_1|$: $\rho_{k+1}/\rho_k \leq |\lambda_2/\lambda_1|$.

Proof. From the definition of ρ_k,

$$\frac{\rho_{k+1}}{\rho_k} = \frac{1}{|\lambda_1|}\sqrt{\frac{\sum_{i\geq 2} a_i^2 \lambda_i^{2k+2}}{\sum_{i\geq 2} a_i^2 \lambda_i^{2k}}}.$$

Put $\omega_i := a_i^2 \lambda_i^{2k} \geq 0$. The fraction under the root is the weighted average $\left(\sum_{i\geq 2}\omega_i\lambda_i^2\right)/\left(\sum_{i\geq 2}\omega_i\right)$ of the numbers $\{\lambda_i^2 : i \geq 2\}$, which cannot exceed their maximum λ_2^2. Therefore

$$\frac{\rho_{k+1}}{\rho_k} \leq \frac{1}{|\lambda_1|}\sqrt{\lambda_2^2} = \left|\frac{\lambda_2}{\lambda_1}\right| < 1.$$

□

Convergence is therefore **geometric** with rate $|\lambda_2/\lambda_1|$. The quantity $|\lambda_1| - |\lambda_2|$ (equivalently the ratio $|\lambda_2/\lambda_1|$) is the **spectral gap**: a large gap means fast, robust convergence; a small gap means slow convergence vulnerable to noise.

2.3 (c) When the spectral gap closes

(i) Take $\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, with $\lambda_1 = 1,\ \lambda_2 = -1$, so $|\lambda_1| = |\lambda_2| = 1$ (no gap). For a generic start $\mathbf{x}_0 = (a,b)$ with $a, b \neq 0$,

$$\mathbf{x}_1 = \mathbf{X}\mathbf{x}_0 = \begin{pmatrix} a \\ -b \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} a \\ b \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} a \\ -b \end{pmatrix}, \ \ldots \quad \Longrightarrow \quad \mathbf{X}^k\mathbf{x}_0 = \begin{pmatrix} a \\ (-1)^k b \end{pmatrix}.$$

The norm $\sqrt{a^2+b^2}$ is preserved, so after normalisation $\mathbf{x}_k$ oscillates forever between the two distinct directions (a,b) and $(a,-b)$ and never converges to a single vector.

(ii) **[EXTRA: explanation not written in the notes.]** A vanishing gap is a *structural* obstruction, not merely slow convergence. The Power Method works by amplifying each eigencomponent by its own factor λ_i^k; the top direction wins only because $|\lambda_1| > |\lambda_i|$. When $|\lambda_1| = |\lambda_2|$ the leading components are amplified by the *same* factor every step, so their relative weights never change: there is no selection mechanism. The iterate keeps whatever mixture it started with – oscillating (as above, when the eigenvalues differ in sign) or, in the degenerate case $\lambda_1 = \lambda_2$, converging to an essentially arbitrary vector of the eigenspace rather than to a unique eigenvector. In the rate of part (b), $|\lambda_2/\lambda_1| = 1$, so ρ_k does not decay at all.

(iii) **[EXTRA: explanation not written in the notes.]** When $|\lambda_2/\lambda_1|$ is close to (but below) 1, two problems compound. First, the number of iterations needed scales like $1/(1-|\lambda_2/\lambda_1|)$, which blows up. Second, each step of double-precision arithmetic injects rounding error of order 10^{-16} into *all* directions, including $\mathbf{v}_2$. The useful per-step separation between $\mathbf{v}_1$ and $\mathbf{v}_2$ is only $\propto |\lambda_2/\lambda_1|^k$; once that signal falls to the level of the accumulated noise, the computed vector drifts within the near-degenerate $\{\mathbf{v}_1, \mathbf{v}_2\}$ subspace and the ordering of the two eigenvalues becomes numerically ambiguous.

2.4 (d) The two symmetric matrices $\mathbf{X}^\top\mathbf{X}$ and $\mathbf{X}\mathbf{X}^\top$

Let $\mathbf{X}$ be a general (rectangular) $m \times n$ matrix.

(i) Symmetric and positive semi-definite.

$$(\mathbf{X}^\top\mathbf{X})^\top = \mathbf{X}^\top(\mathbf{X}^\top)^\top = \mathbf{X}^\top\mathbf{X}, \qquad (\mathbf{X}\mathbf{X}^\top)^\top = \mathbf{X}\mathbf{X}^\top,$$

so both products are symmetric. For positive semi-definiteness, take any real vector $\mathbf{w}$ of the appropriate size:

$$\mathbf{w}^\top(\mathbf{X}^\top\mathbf{X})\mathbf{w} = (\mathbf{X}\mathbf{w})^\top(\mathbf{X}\mathbf{w}) = \|\mathbf{X}\mathbf{w}\|^2 \geq 0, \qquad \mathbf{w}^\top(\mathbf{X}\mathbf{X}^\top)\mathbf{w} = \|\mathbf{X}^\top\mathbf{w}\|^2 \geq 0.$$

[CORRECTED: the notes wrote $\|\mathbf{X}^\top\mathbf{w}\|^2$ for the $\mathbf{X}^\top\mathbf{X}$ case; the correct quantity there is $\|\mathbf{X}\mathbf{w}\|^2$.] Being symmetric (real eigenvalues, by the spectral theorem) and PSD ($\mathbf{w}^\top A\mathbf{w} \geq 0 \Rightarrow$ eigenvalues ≥ 0), both matrices have *real, non-negative* eigenvalues. Hence $\sigma_i := \sqrt{\lambda_i}$ is well defined.

(ii) Same nonzero eigenvalues. Suppose $\mathbf{X}^\top\mathbf{X}\,\mathbf{v} = \lambda\mathbf{v}$ with $\lambda \neq 0$ (so $\mathbf{v} \neq \mathbf{0}$). Left-multiplying by $\mathbf{X}$,

$$\mathbf{X}(\mathbf{X}^\top\mathbf{X}\,\mathbf{v}) = \lambda\,\mathbf{X}\mathbf{v} \quad \Longrightarrow \quad (\mathbf{X}\mathbf{X}^\top)(\mathbf{X}\mathbf{v}) = \lambda\,(\mathbf{X}\mathbf{v}).$$

Moreover $\mathbf{X}\mathbf{v} \neq \mathbf{0}$, since $\mathbf{X}\mathbf{v} = \mathbf{0}$ would give $\mathbf{X}^\top\mathbf{X}\mathbf{v} = \mathbf{0} = \lambda\mathbf{v}$, forcing $\lambda = 0$. Thus $\mathbf{X}\mathbf{v}$ is an eigenvector of $\mathbf{X}\mathbf{X}^\top$ with the same eigenvalue λ. The same argument with $\mathbf{X}^\top$ shows every nonzero eigenvalue of $\mathbf{X}\mathbf{X}^\top$ is an eigenvalue of $\mathbf{X}^\top\mathbf{X}$. Hence $\mathbf{X}^\top\mathbf{X}$ and $\mathbf{X}\mathbf{X}^\top$ have *identical nonzero eigenvalues*, the shared values being σ_i^2.

2.5 (e) Power Method for the SVD: algorithm and worked example

(i) Algorithm (language-agnostic pseudocode). Input: an $N \times M$ matrix $\mathbf{X}$. Output: the top singular triplet $\sigma_1, \mathbf{v}_1, \mathbf{u}_1$.

1. **Form the smaller symmetric matrix.** If $N \geq M$, set $A \leftarrow \mathbf{X}^\top\mathbf{X}$ (size $M \times M$); otherwise set $A \leftarrow \mathbf{X}\mathbf{X}^\top$ (size $N \times N$).
2. **Power iteration on** A**.** Choose a random unit vector $\mathbf{x}$. Repeat $\mathbf{x} \leftarrow A\mathbf{x}/\|A\mathbf{x}\|$ until $\|A\mathbf{x}/\|A\mathbf{x}\| - \mathbf{x}\| <$ tol (or a maximum iteration count).
3. **Read off the singular value.** $\lambda_1 \leftarrow \mathbf{x}^\top A\,\mathbf{x}$ (Rayleigh quotient); $\sigma_1 \leftarrow \sqrt{\lambda_1}$.
4. **Recover the companion vector.** If $A = \mathbf{X}^\top\mathbf{X}$: set $\mathbf{v}_1 \leftarrow \mathbf{x} \in \mathbb{R}^M$ and $\mathbf{u}_1 \leftarrow \mathbf{X}\mathbf{v}_1/\sigma_1 \in \mathbb{R}^N$. If $A = \mathbf{X}\mathbf{X}^\top$: set $\mathbf{u}_1 \leftarrow \mathbf{x} \in \mathbb{R}^N$ and $\mathbf{v}_1 \leftarrow \mathbf{X}^\top\mathbf{u}_1/\sigma_1 \in \mathbb{R}^M$.
5. **Return** $\sigma_1, \mathbf{v}_1, \mathbf{u}_1$.

Choice of matrix. A is square of size $\min(N,M)$. When $N \ll M$ the $N \times N$ matrix $\mathbf{X}\mathbf{X}^\top$ is far cheaper to store and to multiply, and when $M \ll N$ the $M \times M$ matrix $\mathbf{X}^\top\mathbf{X}$ is cheaper – always pick the smaller. **[CORRECTED: the pseudocode text in the notes labelled this rule backwards ("if $N \gg M$ compute $\mathbf{X}\mathbf{X}^\top$"), although the stated reason and the worked example below both use the smaller matrix. The consistent rule is given above.]**

(ii) Worked example by hand.

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \qquad (N = 3,\ M = 2).$$

Here $N > M$, so use the 2×2 matrix

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Eigenvalues: $\det\begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda-1)(\lambda-3) = 0$, so $\lambda = 1$ or $\lambda = 3$; the dominant one is $\lambda_1 = 3$.

Eigenvector for $\lambda_1 = 3$: $(2-3)x + y = 0 \Rightarrow y = x$, giving the unit vector $\mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Singular value: $\sigma_1 = \sqrt{\lambda_1} = \sqrt{3}$.

Left singular vector:

$$\mathbf{u}_1 = \frac{\mathbf{X}\mathbf{v}_1}{\sigma_1} = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Verification.

$$\mathbf{X}\mathbf{v}_1 = \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \qquad \sigma_1\mathbf{u}_1 = \sqrt{3} \cdot \tfrac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \Rightarrow \mathbf{X}\mathbf{v}_1 = \sigma_1\mathbf{u}_1. \checkmark$$

$$\mathbf{X}^\top\mathbf{u}_1 = \tfrac{1}{\sqrt{6}}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \tfrac{1}{\sqrt{6}}\begin{pmatrix} 3 \\ 3 \end{pmatrix} = \tfrac{3}{\sqrt{6}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \sigma_1\mathbf{v}_1 = \sqrt{3} \cdot \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \sqrt{\tfrac{3}{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

and $\frac{3}{\sqrt{6}} = \sqrt{\frac{3}{2}}$, so $\mathbf{X}^\top\mathbf{u}_1 = \sigma_1\mathbf{v}_1$. ✓Both SVD relations hold, confirming $(\sigma_1, \mathbf{v}_1, \mathbf{u}_1)$.